

Precision Machining Caliper – Back – Setup and Planning

Course Challenge: Review a 3D model, plan tools and toolpaths, and create a CAM setup for the first operation.

Criteria: During the course, we walked through the process to plan, setup, and verify a program for the brake caliper piston. In this challenge, you will take the back portion of the caliper, review the blueprint and model to identify tools required to machine the part, identify how to hold the part, and create a new CAM setup for only the caliper back. The provided dataset will include the entire caliper assembly as well as a dataset for a vise with hard jaws

Supplied datasets are **05-0404 – Vise with Hard Jaws.f3d**, **Precision Machining Caliper – INCH.f3d**, and **Precision Machining Caliper – INCH Drawing Rev B.pdf**. Explore creating a Derive of the back caliper into its own design and inserting the vise model into that design. This will be the basis of a new CAM program. Once there is a design created with a vise and the rear portion of the caliper, navigate to the Manufacture workspace, and create a new CAM setup. It is suggested that the inside of the caliper be machined first, but it is not required.

The following steps are required:

- **Dataset Setup** – Create a Derive of the caliper back and set it up in the supplied vise.
- **Tool Library Setup** – List tools you plan to use on the caliper and search the Fusion 360 Samples library for applicable tools.
- **CAM Setup** – Create a Setup for machining OP1.

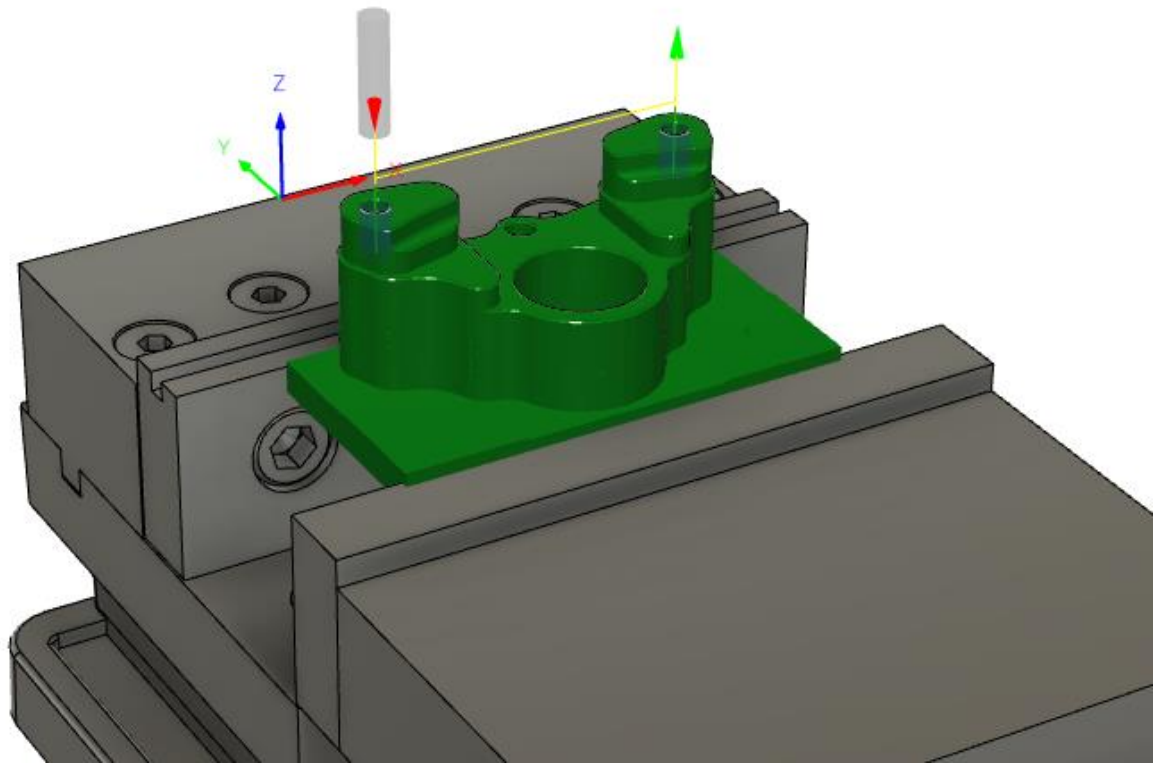


Figure 1: OP1